# Natural Language Processing

## Hard & Soft-Label for Hate Speech Classification

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# The dataset (Davidson et al.)

## Annotation Guidelines

CrowdFlower workers labeled tweets into three classes:

- **Hate Speech (0):** Language intended to derogate, humiliate, or insult a *targeted group*.
- **Offensive (1):** Contains slurs/profanity but does *not* target specific groups.
- **Neither (2):** No offensive language.

## Dataset Profile

Total tweets analyzed: **24,783**

- Heavily skewed toward the "Offensive" class (**19,190** tweets).
- "Hate" is a severe minority class (only **1,430** tweets).
- Stratified split: **70%** Train / **15%** Val / **15%** Test.

### Class Examples

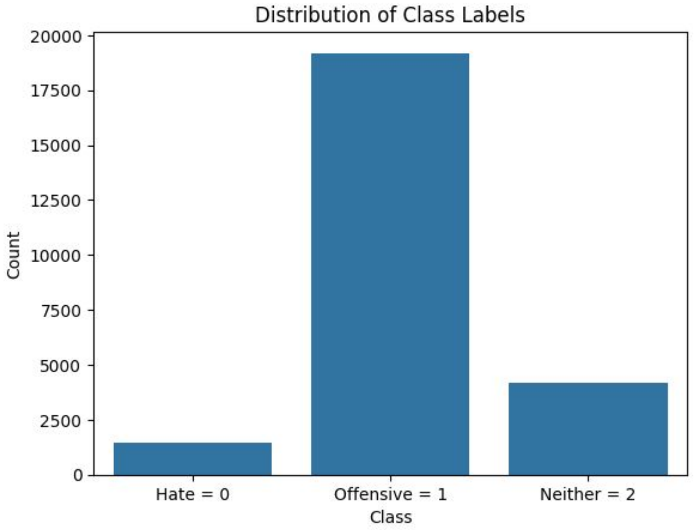
**Hate:** *"@denytheprophecy thanks for ignoring my texts fag"*
**Offensive:** *"@MvckFadden: "Stay beautiful you bitch""*
**Neither:** *"I just want vanilla Oreos"*

# The Problems

**Task:** Distinguishing genuine Hate Speech from slang and offensive language.

**Lexical Challenge:** Severe vocabulary overlap. Slurs are heavily utilized in both Hate and Offensive classes.

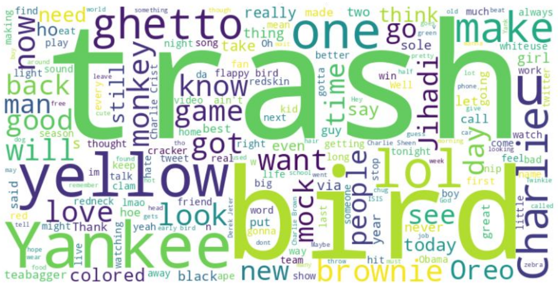
**Data Distribution:** We want to predict the minority class.




**Hate Speech (Class 0)**

**Offensive Language (Class 1)**

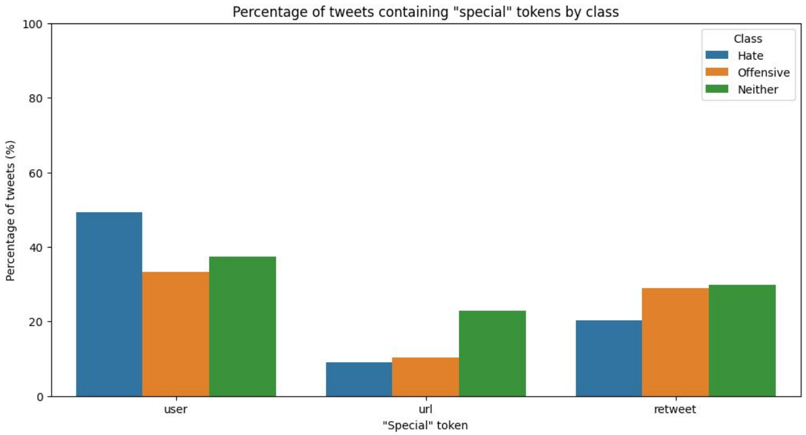
**Neither (Class 2)**



# General Preprocessing Pipeline

To preserve tweet context, specific substitutions were made during cleaning:

- **Retweets:** `RT @handle:` was replaced with **`retweet`**
- **User Mentions:** Twitter handles (e.g., `@mayasolovely`) became **`user`**
- **Links:** URLs were replaced with the token **`url`**




### Raw Input Example

```
“@ClicquotSuave: LMAOOOOOOOOOO
this nigga @Krillz Nuh Care
http://t.co/AAnpSUjmYI”
<bitch want likes for some depressing
shit..foh
```

### Normalized Output

```
user LMAOOOOOOOOOO this nigga user
url bitch want likes for some depressing
shit..foh
```

# Baseline 1: TF-IDF + Logistic Regression

## Preprocessing Pipeline

- Lowercase
- Word-level tokenization
- Punctuation removal
- English stopword exclusion
- Extraction of both Bigrams & Trigrams

### Processing Example

Raw input sentence:

`"Just trynna have a bad bitch come thru 😴"`

Resulting tokens:

`['trynna', 'bad', 'bitch', 'come', 'thru']`

| Class | Precision | Recall | F1-Score (95% CI) |
|---|---|---|---|
| 0: Hate | 0.294 | 0.640 | 0.403 (0.356 - 0.449) |
| 1: Offensive | 0.972 | 0.833 | 0.897 (0.888 - 0.906) |
| 2: Neither | 0.745 | 0.934 | 0.829 (0.807 - 0.849) |
| **macro avg** | **0.67** | **0.80** | **0.71** |

# Error Analysis for TF-IDF

### Top Words: HATE

| WORD | COEF |
|---|---|
| faggot | 6.840 |
| nigger | 6.531 |
| nigga | 6.042 |
| fag | 4.936 |
| white_trash | 4.567 |
| dyke | 3.945 |
| queer | 3.595 |
| spic | 3.496 |
| fuck | 3.361 |
| wetback | 3.279 |

### Top Words: OFFENSIVE

| WORD | COEF |
|---|---|
| bitch | 12.271 |
| hoe | 5.965 |
| pussy | 5.886 |
| shit | 3.042 |
| eat_pussy | 2.146 |
| hoe_ai_loyal | 2.141 |
| gon | 1.734 |
| fuck | 1.638 |
| ya | 1.620 |
| as | 1.612 |

### Top Words: NEITHER

| WORD | COEF |
|---|---|
| bird | 4.194 |
| yankee | 3.689 |
| charlie | 3.371 |
| yellow | 3.170 |
| mock | 2.854 |
| brownie | 2.797 |
| oreo | 2.491 |
| trash | 2.477 |
| jihadi | 2.261 |
| colored | 2.149 |

### False Positives Examples

*"Because of the message that sends to LGBTQ residents in the state. Not as if West Virginia is a paradise for queer people. Today is major"*
*"user Would you still love me when I'm no longer young and beautiful?" No. Ugly monkey ass"*
*“user god doesn't send retards to hell"*
*“Photo: Giving you that trailer park trash. #transformthursday #ladykimora #vegasqueens #vegasshowgirls url”*
*'Im the biggest pussy when it comes to scary stuff'*

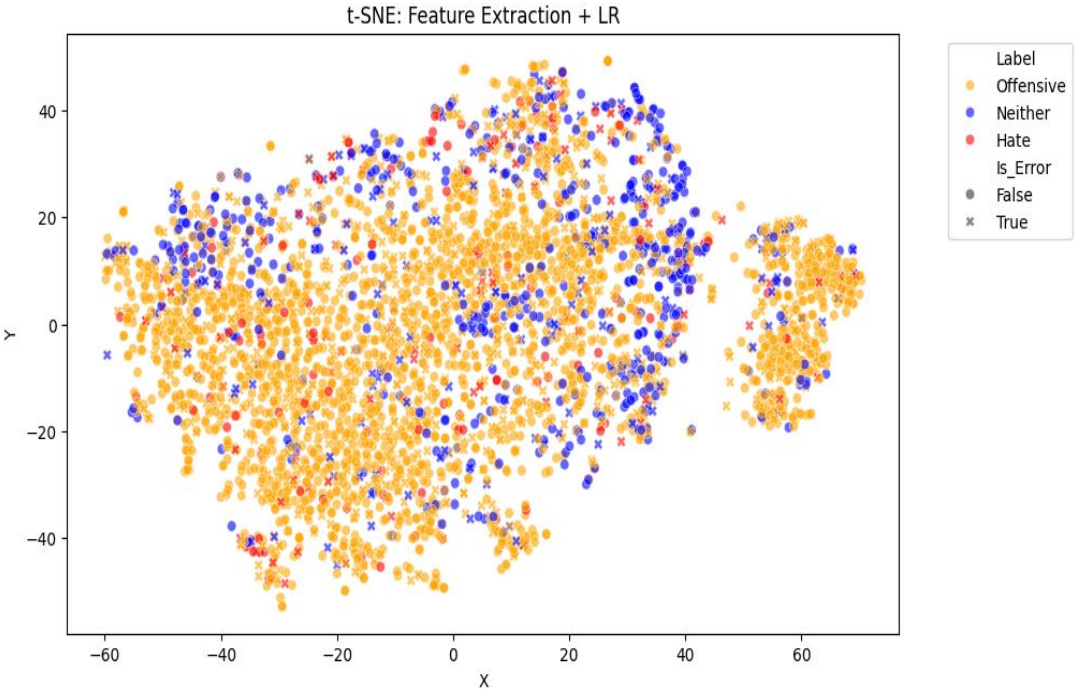
# Baseline 2: Frozen BERT Feature Extraction

## [CLS] EMBEDDING ANALYSIS

Extraction of 768-dimensional vectors from the [CLS] token of bert-base-cased.

The modest results stem from the inability of general linguistic structures (Wikipedia/BookCorpus) to decode internet slang without domain-specific fine-tuning.

| Classe | Precision | Recall | F1-Score (95% CI) |
|---|---|---|---|
| 0: Hate | 0.186 | 0.565 | 0.279 (0.240 - 0.318) |
| 1: Offensive. | 0.945 | 0.746 | 0.834 (0.823 - 0.845) |
| 2: Neither | 0.590 | 0.751 | 0.661 (0.631 - 0.689) |
| **macro avg** | **0.57** | **0.69** | **0.59** |



# Domain Adaptation: LoRA vs. Full Fine-Tuning

Both models were trained with Cross Entropy and by applying weights: [5.7766, 0.4305, 1.9843]

## BERT LoRA

Rank: 16, Alpha: 32, Dropout: 0.1, LR: 1e-4, Epochs: 5, Weight Decay: 0.01

Optimizes only internal transformer projection matrices (`query`, `key`, `value`) for parameter-efficient tuning.

| Class | Precision | Recall | F1-Score (95% CI) |
|---|---|---|---|
| 0: Hate | 0.287 | 0.682 | 0.404 (0.359 - 0.449) |
| 1: Offensive | 0.972 | 0.852 | 0.908 (0.900 - 0.916) |
| 2: Neither | 0.841 | 0.922 | 0.879 (0.860 - 0.897) |
| **macro avg** | **0.70** | **0.82** | **0.73** |

## BERT Full Fine-Tuning

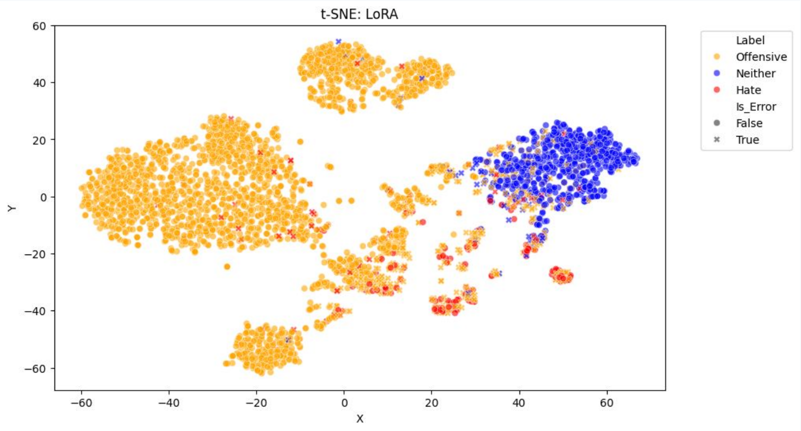
LR: 2e-5, Epochs: 5, Weight Decay: 0.05

Updates all model parameters. Maximizes domain adaptation capacity but requires significantly more training compute.

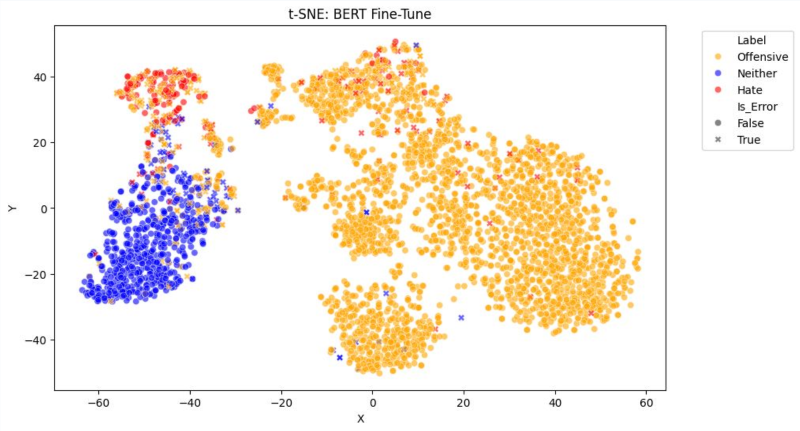
| Class | Precision | Recall | F1-Score (95% CI) |
|---|---|---|---|
| 0: Hate | 0.402 | 0.594 | 0.479 (0.426 - 0.530) |
| 1: Offensive | 0.957 | 0.921 | 0.939 (0.933 - 0.945) |
| 2: Neither | 0.880 | 0.890 | 0.885 (0.865 - 0.902) |
| **macro avg** | **0.75** | **0.80** | **0.77** |

# LoRA and Full Fine-Tuning t-SNE

## BERT LoRA t-SNE




## BERT Full Fine-Tuning t-SNE

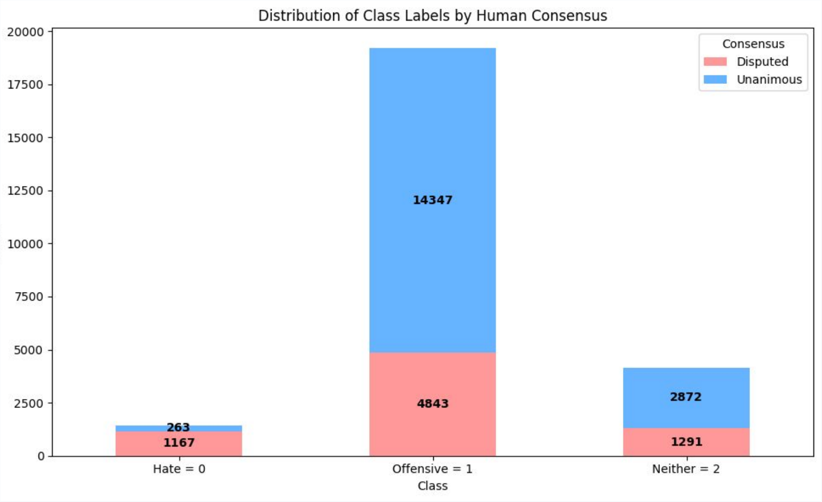


# Are Humans Even Certain?

Analyzing raw CrowdFlower annotator votes reveals a flaw in the ground truth: the vast majority of "Hate" instances are **Disputed** (mixed human votes) .

Forcing a deterministic "Hard Label" via majority vote destroys crucial information regarding linguistic ambiguity.

### Distribution of Class Labels




### Accuracy by Consensus Type (on Test)

unanimous_hate: 85.37% (n=41)

unanimous_offensive: 96.67% (n=2164)

unanimous_neither: 96.00% (n=425)

### Accuracy Disputed Sub-classes (on test)

disputed-hate: 53.18% (n=173)

disputed-offensive: 78.32% (n=715)

disputed-neither: 74.00% (n=200)

# Soft Labels

Shift from predicting a single majority class to predicting the **distribution of human votes**.

**Example 1**

*"bitch who do you love"*

Human vote: `[1, 2, 0]`

`Hard Label: 1`

Soft Label: `[0.333, 0.666, 0.0]`

**Example 2**

*"is that ya bitch"*

Human vote: `[0, 3, 0]`

`Hard Label: 1`

Soft Label: `[0.0, 1.0, 0.0]`

**Optimization Objective**

Using cross entropy, as was done with hard labels, and NOT using any weights, **it has to minimize the loss by perfectly replicating the exact proportion of human disagreement**.

# Soft Label Training Results

Optimizing the classification threshold to 0.34 balances precision and recall for the minority class.

## Initial Soft Labels

Default Rule: Highest Probability

Standard evaluation shows lower performance on class **Hate** due to the default decision rule.

| Class | Precision | Recall | F1-Score |
|---|---|---|---|
| 0: Hate | 0.530 | 0.420 | **0.470** |
| 1: Offensive | 0.940 | 0.960 | **0.950** |
| 2: Neither | 0.890 | 0.900 | **0.900** |
| **marco avg** | 0.79 | 0.76 | **0.77** |

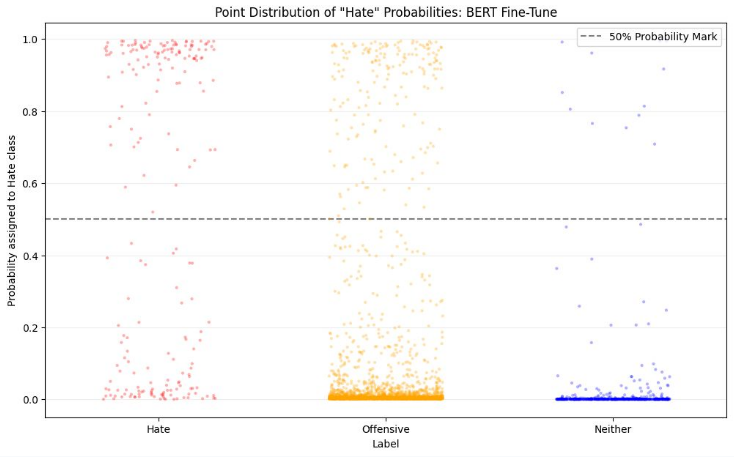
## Optimized Soft Labels

**Optimal F1-Hate Threshold: 0.34**
(Validation Hate F1: 0.4879)

| Class | Precision | Recall | F1 (95% CI) |
|---|---|---|---|
| 0: Hate | 0.441 | 0.590 | **0.504** (0.449-0.555) |
| 1: Offensive | 0.959 | 0.932 | **0.946** (0.939-0.951) |
| 2: Neither | 0.889 | 0.902 | **0.896** (0.878-0.913) |
| **marco avg** | 0.76 | 0.81 | **0.78** |

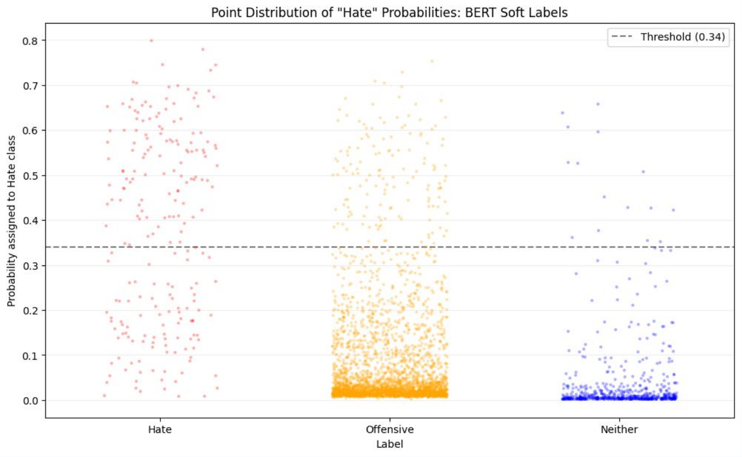
# Why optimize the threshold only with soft labels?

## Hard Labels (Argmax Rule)


Soft labels reduced
Offensive → Hate
false positives

## Soft Labels (Optimized 0.34 Threshold)

# Did the model learn human uncertainty?

## DISTRIBUTION DISTANCE ANALYSIS

| Consensus Type | Avg KL Div | Avg MAE |
|---|---|---|
| unanimous_hate | 0.9387 | 35.20% |
| unanimous_off | 0.0972 | 5.34% |
| unanimous_neither | 0.1359 | 6.97% |

### Disputed Sub-classes

| | | |
|---|---|---|
| disputed-hate | 0.4661 | 21.42% |
| disputed-offensive | 0.3748 | 15.38% |
| disputed-neither | 0.4849 | 19.05% |

### "BEST" PREDICTION

**KL Div: 0.0021**

*"Fuck the church you attend , fuck your god, fuck the nigguh you call the savior . It's a fucking new world order baby"*

**Human Votes** [0.33, 0.67, 0.00]

**BERT Distribution** [0.34, 0.66, 0.00] Avg Error/Class: 0.3%

### WORST PREDICTION

**KL Div: 2.8727**

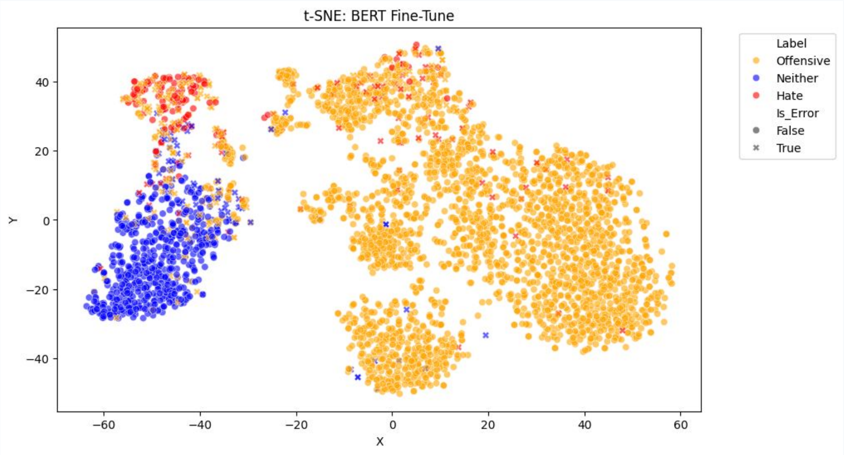
*"user user user once f*g." You got pressed to the ground? ..hang up your cleats n never put them back on..."*

**Human Votes** [0.00, 0.33, 0.67]

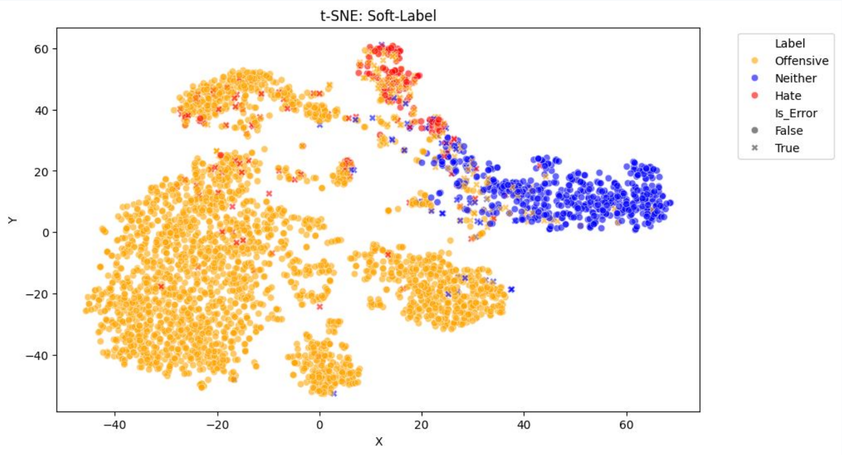
**BERT Distribution** [0.44, 0.55, 0.01] Avg Error/Class: 44.0%

# t-SNE: Hard-Labels vs Soft-Labels

## Hard-Labels




## Soft-Labels



# Explainability: Best prediction case

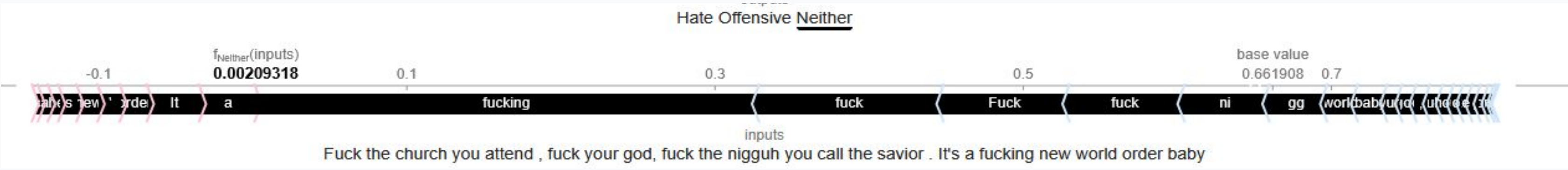
**Human Votes**

Hate: 1 | Offensive: 2 | Neither: 0

**BERT Soft Probabilities**

Hate: 0.34 | Offensive: 0.66 | Neither: 0.00





# Explainability: Worst prediction case

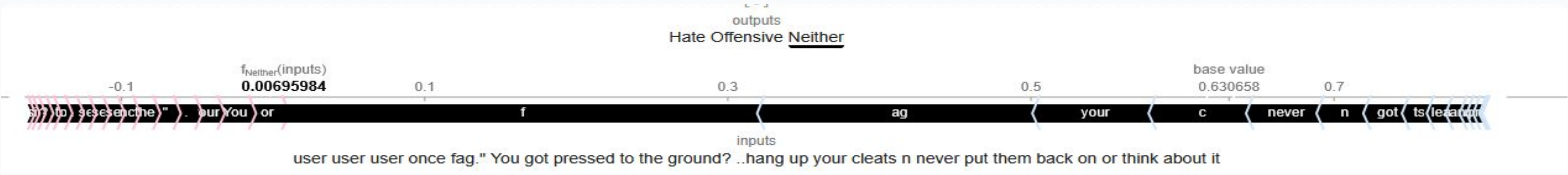
**HUMAN VOTES**

Hate: 0 | Offensive: 1 | Neither: 2

**BERT SOFT PROBABILITIES**

Hate: 0.44 | Offensive: 0.55 | Neither: 0.01





# Explainability: Another worst prediction case

**Human Votes**

Hate: 0 | Offensive: 1 | Neither: 2

**BERT Soft Probabilities**

Hate: 0.42 | Offensive: 0.57 | Neither: 0.01



# BERT: Cased vs Uncased

## CASED

| Class | Hard F1 (CI) | Soft F1 (CI) |
|---|---|---|
| Hate | 0.48 (0.43 - 0.53) | **0.50** (0.45 - 0.55) |
| Offensive | 0.94 (0.93 - 0.95) | **0.95** (0.94 - 0.95) |
| Neither | 0.88 (0.87 - 0.90) | **0.90** (0.88 - 0.91) |

## UNCASED

| Class | Hard F1 (CI) | Soft F1 (CI) |
|---|---|---|
| Hate | 0.47 (0.41 - 0.52) | **0.51** (0.45 - 0.56) |
| Offensive | 0.94 (0.93 - 0.94) | **0.95** (0.94 - 0.96) |
| Neither | 0.90 (0.88 - 0.92) | 0.90 (0.88 - 0.91) |

# Conclusion & Future Work

## Conclusion

**Modeling subjective human language requires probabilistic representations of uncertainty rather than rigid categorical labels.**

Soft Labels resolve overconfidence and improve decision boundaries.

## Future Work

- **Domain-Specific Tokenization:** Explore custom tokenization strategies tailored for niche domains.
- **Demographic-Rich Pre-training:** Pre-train on corpora rich in demographic and regional slang to effectively decode colloquial internet jargon.

# Thank you
# for
# your attention